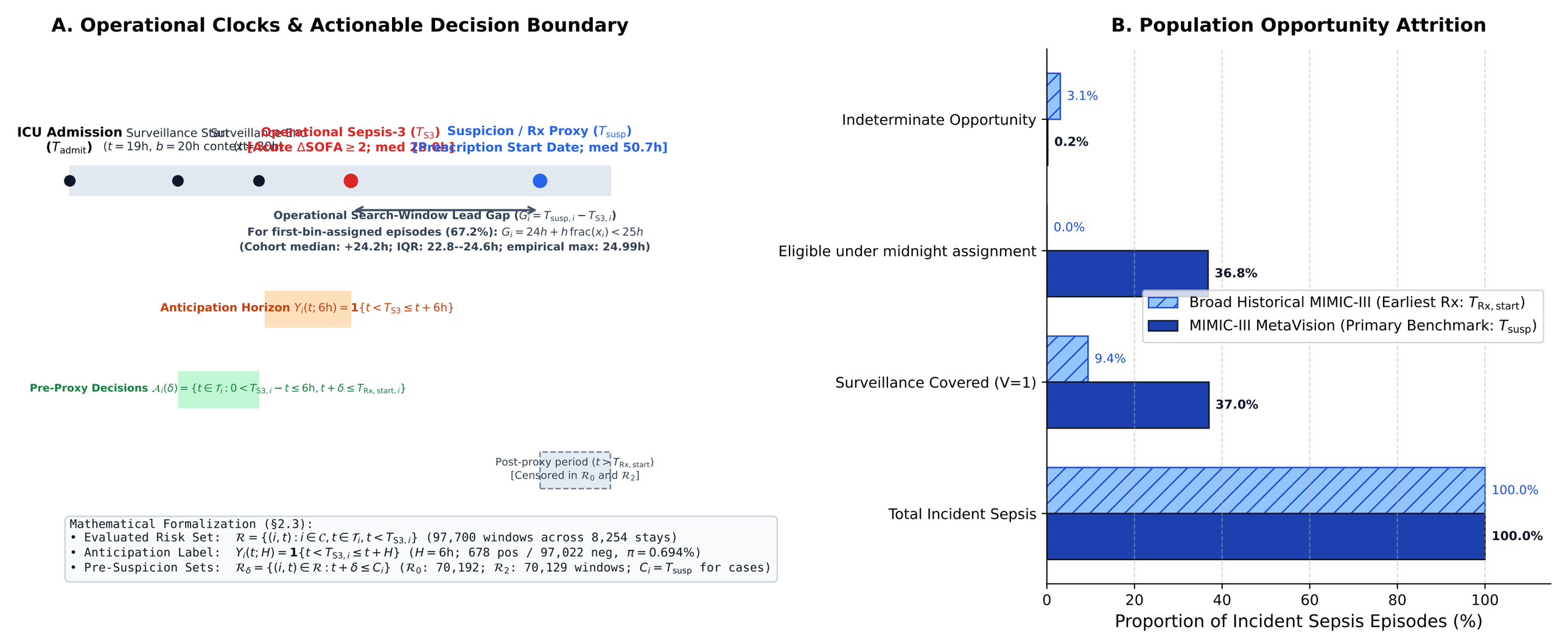

A. Operational Clocks & Actionable Decision Boundary
ICU Admission (T_{admit})
Surveillance Start ($t = 19$h, $b = 20$h context)
Surveillance End ($t + 30$h)
Operational Sepsis-3 (T_{S3}) [Acute ΔSOFA ≥ 2; med 29.0h]
Suspicion / Rx Proxy (T_{susp}) [Prescription Start Date; med 50.7h]
Operational Search-Window Lead Gap ($G_i = T_{susp,i} - T_{S3,i}$)
For first-bin-assigned episodes (67.2%): $G_i = 24h + h\,\mathrm{frac}(x_i) < 25h$
(Cohort median: +24.2h; IQR: 22.8--24.6h; empirical max: 24.99h)
Anticipation Horizon $Y_i(t; 6\mathrm{h}) = \mathbf{1}\{t < T_{S3} \le t + 6\mathrm{h}\}$
Pre-Proxy Decisions $\mathcal{A}_i(\delta) = \{t \in \mathcal{T}_i : 0 < T_{S3,i} - t \le 6\mathrm{h}, t + \delta \le T_{\mathrm{Rx,start},i}\}$
Post-proxy period ($t > T_{\mathrm{Rx,start}}$)
[Censored in $\mathcal{R}_0$ and $\mathcal{R}_2$]
Mathematical Formalization (§2.3):
• Evaluated Risk Set: $\mathcal{R} = \{(i,t) : i \in \mathcal{C}, t \in \mathcal{T}_i, t < T_{S3,i}\}$ (97,700 windows across 8,254 stays)
• Anticipation Label: $Y_i(t;H) = \mathbf{1}\{t < T_{S3,i} \le t + H\}$ ($H = 6$h; 678 pos / 97,022 neg, $\pi = 0.694\%$)
• Pre-Suspicion Sets: $\mathcal{R}_\delta = \{(i,t) \in \mathcal{R} : t + \delta \le C_i\}$ ($\mathcal{R}_0$: 70,192; $\mathcal{R}_2$: 70,129 windows; $C_i = T_{susp}$ for cases)
B. Population Opportunity Attrition
Indeterminate Opportunity
Eligible under midnight assignment
Surveillance Covered (V=1)
Total Incident Sepsis
3.1%
0.2%
0.0%
36.8%
9.4%
37.0%
100.0%
100.0%
Broad Historical MIMIC-III (Earliest Rx: $T_{\mathrm{Rx,start}}$)
MIMIC-III MetaVision (Primary Benchmark: T_{susp})
0
20
40
60
80
100
Proportion of Incident Sepsis Episodes (%)